

Synopsis: Kinematics is the study of motion using physics concepts and equations.

- I. What is motion?
- II. How do we describe motion in physics?
- III. How do we apply the ideas to real situations?
- IV. Three key equations in Kinematics

In Calculus Notation

Velocity is the derivative of position, and acceleration is the derivative of velocity (the second derivative of position):

$$v = dx/dt \quad a = dv/dt = d^2x/dt^2$$

When acceleration is constant, integrating $dv/dt = a$ once gives $v = v_o + at$ (Eq. 1), and integrating again gives $\Delta x = v_o t + \frac{1}{2}at^2$ (Eq. 2) — the same two equations above, arrived at directly from calculus. Show the integration steps: